Bracketing a root by bisection

Upper high school · analysis and algorithms · Worked solutions · 12

Given information. For exercise 1: seek a zero of $f(x) = x^2 - 2$ on $[1, 2]$. After two halvings, the retained interval is $[1.25, 1.5]$.

Exercise 1. Carry out a third step of the example.

Solution. The midpoint is 1.375, whose square is $1.890625 < 2$. The new interval is $[1.375, 1.5]$.

Exercise 2. Starting with width 1, how many halvings guarantee width at most $\frac{1}{1000}$?

Solution. $2^9 = 512 < 1000 \leq 1024 = 2^{10}$, so 10 halvings suffice and 9 do not.

Exercise 3. Why does the sign change of $f(x) = \frac{1}{x}$ between -1 and 1 not prove the existence of a zero?

Solution. The function is undefined at 0 and is not continuous on $[-1, 1]$. The theorem's hypothesis fails; $\frac{1}{x}$ is never zero.

Exercise 4. A root lies in $[1.4, 1.5]$. What maximum error is guaranteed by midpoint 1.45?

Solution. The half-width is $\frac{1.5-1.4}{2} = 0.05$.

Problem · Investigate, justify, conclude. For $f(x) = x^3 - x - 1$ on $[1, 2]$, prove existence and uniqueness of a root α. Carry out three bisection steps. How many halvings guarantee that the final midpoint approximates α with error at most 0.001?

Solution. $f(1) = -1$ and $f(2) = 5$, and f is continuous. Also $f'(x) = 3x^2 - 1 \geq 2$ on $[1, 2]$, so f is strictly increasing. Midpoints 1.5, 1.25, 1.375 give $[1, 1.5]$, $[1.25, 1.5]$, $[1.25, 1.375]$ because f takes values 0.875, -0.296875 and 0.224609375. After n halvings, midpoint error is at most $\frac{1}{2^{n+1}}$. Thus $2^{n+1} \geq 1000$, requiring $n \geq 9$.